

Full production three-component gradient flow: nonlinear-map audit

TECT verification-first repository · claim A2-FULL-PRODUCTION-WELLPOSED

first issued 2026-07-17 · this version issued 2026-07-17 · v1.1

1. Purpose and scope

This v1.1 note closes only the first analytic audit of the full-production P2 route. It does not change the older scalar claim **A2-PDE-WELLPOSED**, and it does not promote the full claim beyond T4. The scope remains the real- L^2 gradient flow of the canonical P1 reference functional on the fixed three-torus, with three complex components, positive regularisers, and $\eta_{\text{shell}} = 0$. The historical backend remains a non-variational proxy and is excluded.

The purpose is precise: expand the Class-II Euler–Lagrange map in six real coordinates, prove that it has spatial order at most two, and establish the local $H^2 \rightarrow L^2$ Lipschitz estimate required by the fourth-order semigroup local-existence route. The Galerkin energy passage and the positive-time smoothing bootstrap are not addressed here; their two gates remain open.

2. Six-real-coordinate Class-II density

Write

$$u = (\text{Re } \Psi_1, \text{Re } \Psi_2, \text{Re } \Psi_3, \text{Im } \Psi_1, \text{Im } \Psi_2, \text{Im } \Psi_3) \in \mathbb{R}^6.$$

For each P1 Hermitian generator T_A , let A_A denote its real symmetric realification. Then

$$\rho = u^T u, \quad m_A = u^T A_A u, \quad q_A = \frac{m_A}{\rho + \epsilon_\rho}.$$

With $w_j = \partial_j u$, define

$$p_A(u) = 2A_A u, \quad s_A(u) = 2(A_A - q_A(u)I_6)u.$$

The identities

$$\partial_j m_A = p_A(u)^T w_j, \quad K_{A,j} = s_A(u)^T w_j$$

turn the full Class-II density into

$$G_{II}(u, \nabla u) = \frac{1}{2} \sum_{j=1}^3 w_j^T B(u) w_j,$$

where

$$B(u) = \sum_A \left[a p_A p_A^T + b(p_A s_A^T + s_A p_A^T) + c s_A s_A^T \right]. \quad (2.1)$$

Here a, b, c are exactly the production coefficients derived from the P1 manifest. The matrix is symmetric. Its definition is an identity, rather than an approximation or an expansion around a nonzero background.

The regulariser is decisive: $\epsilon_\rho > 0$ makes q_A , hence B , a C^∞ function on all of $\mathbb{R}^6$. Moreover

$$|q_A(u)| \leq \|A_A\|_{\text{op}} \frac{\rho}{\rho + \epsilon_\rho} \leq \|A_A\|_{\text{op}}. \quad (2.2)$$

For every finite R , the functions B , DB , and D^2B have finite supremum norms on $\{|u| \leq R\}$. No inverse power of ρ is present without the positive ϵ_ρ denominator.

3. Explicit Euler–Lagrange expansion

For a density of the form (2.1), componentwise differentiation gives

$$(N_{II}(u))_\gamma = -B_{\gamma\beta}(u)\Delta u_\beta + \sum_{j=1}^3 \left[\frac{1}{2} \partial_\gamma B_{\alpha\beta}(u) - \partial_\alpha B_{\gamma\beta}(u) \right] (\partial_j u_\alpha)(\partial_j u_\beta), \quad (3.1)$$

with repeated real-coordinate indices summed from 1 to 6. Thus there are only two types of terms:

$$B(u)\nabla^2 u, \quad C(u)[\nabla u, \nabla u], \quad (3.2)$$

where C is a smooth tensor built from DB . In particular, Class-II contributes no spatial derivative of order three or four. Since the pinned Brazovskii coefficient is $Y = 1 > 0$, the fourth-order linear operator remains the unique principal order-four part of the PDE.

For auditability, the derivative used in (3.1) is completely explicit. For a direction $h \in \mathbb{R}^6$,

$$Dq_A(u)[h] = \frac{2(h^T A_A u)(\rho + \epsilon_\rho) - 2m_A(h^T u)}{(\rho + \epsilon_\rho)^2}, \quad (3.3)$$

$$Dp_A(u)[h] = 2A_A h, \quad Ds_A(u)[h] = 2(A_A - q_A I_6)h - 2Dq_A(u)[h]u. \quad (3.4)$$

Applying the product rule to (2.1) gives $DB(u)[h]$ with no hidden differential operator. Equations (3.1)–(3.4) are the requested termwise real-coordinate expansion.

4. Local H2-to-L2 Lipschitz estimate

Fix an H^2 ball of radius R on the fixed three-torus. Sobolev embedding gives

$$H^2 \hookrightarrow L^\infty, \quad H^2 \hookrightarrow W^{1,6}.$$

Equation (2.2) and the bounded-set smoothness of B, DB, D^2B give constants C_R for all coefficient differences. From (3.2),

$$\begin{aligned} \|B(u)\Delta u - B(v)\Delta v\|_{L^2} &\leq C_R \|u - v\|_{H^2}, \\ \|C(u)[\nabla u, \nabla u] - C(v)[\nabla v, \nabla v]\|_{L^2} &\leq C_R \|u - v\|_{H^2}. \end{aligned} \quad (4.1)$$

For the second line, each product of two first derivatives is in L^3 by $L^6 \cdot L^6 \subset L^3$, and $L^3 \hookrightarrow L^2$ on the fixed finite-volume torus. The differences split into one derivative difference and one bounded $W^{1,6}$ factor, plus a coefficient difference controlled by $\|u - v\|_{L^\infty}$. This proves

$$\|N_{II}(u) - N_{II}(v)\|_{L^2} \leq C_R \|u - v\|_{H^2}. \quad (4.2)$$

The local potential has real gradient

$$N_{\text{pot}}(u) = \lambda \rho u + \gamma \rho^2 u,$$

and obeys the same estimate because $H^2(\mathbb{T}^3)$ is an algebra. The family and lock terms are bounded linear maps. Hence the full lower-order map $N : H^2 \rightarrow L^2$ is locally Lipschitz in the pinned canonical scope.

This closes the mapping input for the standard analytic-semigroup local solution construction. It does not itself establish the later Galerkin chain rule, global continuation, continuous dependence, or smoothing claims.

5. Independent executable audit

The independent NumPy script `codes/foundations/a2_full_production_nonlinear_mapping_audit.py` does not call the Torch-autodiff P1 backend. It reads the P1 manifest and verifies the backend source hash, then checks the six-real-coordinate identities against central finite differences of local jets. The recorded run reports 14/14 assertions passing:

$$\begin{aligned}\max |m_{\mathbb{C}} - m_{\mathbb{R}}| &= 8.881784197001252 \times 10^{-16}, \\ \max \text{relerr}(Dq) &= 2.70426292470205 \times 10^{-10}, \\ \max \text{relerr}(DB) &= 1.1811986217680235 \times 10^{-10}, \\ \max \text{relerr}(E_{II}) &= 9.042801480415788 \times 10^{-11}.\end{aligned}$$

The complex-to-real density identity has relative error $1.1102230246251565 \times 10^{-16}$. These are reproducibility checks of the expansion and its factors; the analytic proof of (4.2), not the float values, is the closure basis.

6. Devil’s-advocate

- “ K_A divides by ρ , so the zero field is singular.” DISMISSED in the stated scope: the pinned functional has $\rho + \epsilon_\rho$ with $\epsilon_\rho = 10^{-12} > 0$; (3.3) is defined at $u = 0$ and the executable includes that case.
- “The $C(u)[\nabla u, \nabla u]$ term loses one derivative.” DISMISSED: $H^2 \hookrightarrow W^{1,6}$ makes the product L^3 , and on the fixed torus $L^3 \hookrightarrow L^2$. The difference estimate needs only $D^2 B$ on an L^∞ -bounded ball.
- “Realification may hide a factor of two or the complex cross-term.” DISMISSED by the independent complex-to-real density identity and local Euler–Lagrange jet test, both of which pass with the recorded errors above.
- “This proves global well-posedness.” UPHELD as an invalid reading. The Galerkin chain rule/energy passage and smoothing bootstrap are still separate open gates.
- “The conclusion transfers to the historical solver.” UPHELD as an invalid reading. The audit is only for the canonical P1 variational functional; the historical backend remains explicitly non-variational.

7. Result footer

Result ID:	A2-FULL-PRODUCTION-WELLPOSED
Precise statement:	The full canonical lower-order Euler--Lagrange map, including Class-II, is locally Lipschitz $H^2 \rightarrow L^2$.
Scope:	Fixed T3, six-real-coordinate C3 field, canonical P1 functional, positive regularisers, <code>eta_shell=0</code> .
Dependencies:	A1-PRODUCTION-FUNCTIONAL-REALISATION.
Evidence grade:	ANALYTIC + EXECUTED; T4 proof candidate.
Reproduction command:	<code>python codes/foundations/a2_full_production_nonlinear_mapping_audit.py</code>
Expected output:	14/14 PASS; A2-FULL-NONLINEAR-MAPPING-AUDIT-PASS; exit 0.
Falsification gate:	A singular coefficient, a Class-II derivative above order two, or a failed bounded-ball $H^2 \rightarrow L^2$ bound.
Tier before / after:	T4 / T4 (one analytic gate discharged only).
No-overclaim statement:	Not energy continuation, global PDE closure, smoothing, T6, T7, historical proxy, or <code>eta_shell!=0</code> .
Next required action:	Audit the Galerkin chain rule, energy identity, and coercive continuation in A2-FULL-ENERGY-CONTINUATION-AUDIT.